Answers: Introduction to t-testingg

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Summary

Answers to a selection of questions for the study guide on introduction to t-testing.

These are the answers to [*Questions: Introduction to t*-testing].

Please attempt the questions before reading these answers.

Q1

1.1. Sample size less than 30.

1.2. A hypothesis test where the test statistic follows the t distribution.

1.3. A t-test with equal variances.

1.4. A t-test with unequal variances.

Q2

2.1. $\bar{x}$ is the sample mean. μ_0 is the known value. s is the sample standard deviation. n is the sample size.

2.2. μ_1 is sample mean 1. μ_2 is sample mean 2. s_1 is sample standard deviation 1. s_2 is sample standard deviation 2. n_1 is sample size 1. n_2 is sample size 2.

2.3.

a) $s_1 = s_2$

b) $s_1 \neq s_2$

Q3

3. 1. $H_0 : \bar{x} = \mu_0 H_1 : \bar{x} > \mu_0$

3.2. $H_0 : \mu_1 = \mu_2 H_1 : \mu_1 > \mu_2$

3.3. $H_0 : \mu_1 = \mu_2 H_1 : \mu_1 \neq \mu_2$

$(10 - 7)/(s/\sqrt{19}) = 26.1534$

Student's t-test.

One sample t-test are always Student t-tests.

$(12 - 13)/((2/\sqrt{15}) + (2/\sqrt{16})) = -0.9839$

Student's t-test.

$s1_{=}s_2$

$(45 - 51)/((3/\sqrt{19}) + (3.5/\sqrt{18})) = -3.9651$

Welch's t-test.

$s_1 \neq s_2$

Version history and licensing

v1.0: initial version created 03/26 by Millie Harris (as part of a University of St Andrews VIP project)